Linear Regression Analysis: Aligning Host Pricing Decisions with Consumer Preferences in Amsterdam, North Holland, Netherlands

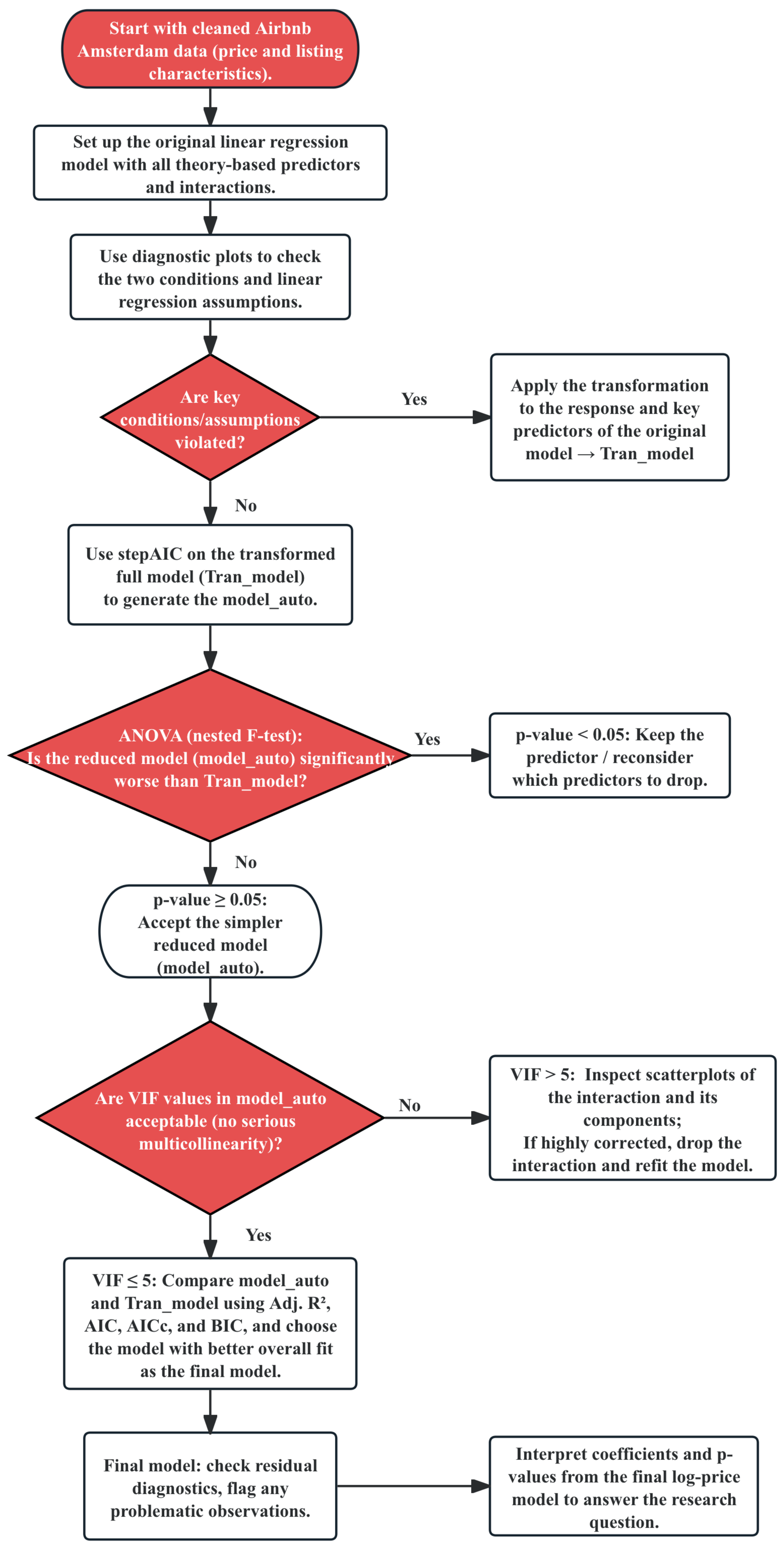
Introduction

Research Question: To what extent do Airbnb hosts' pricing decisions reflect consumer preferences, as represented by host attributes, customer review metrics, and different property features, in Amsterdam, North Holland, Netherlands?

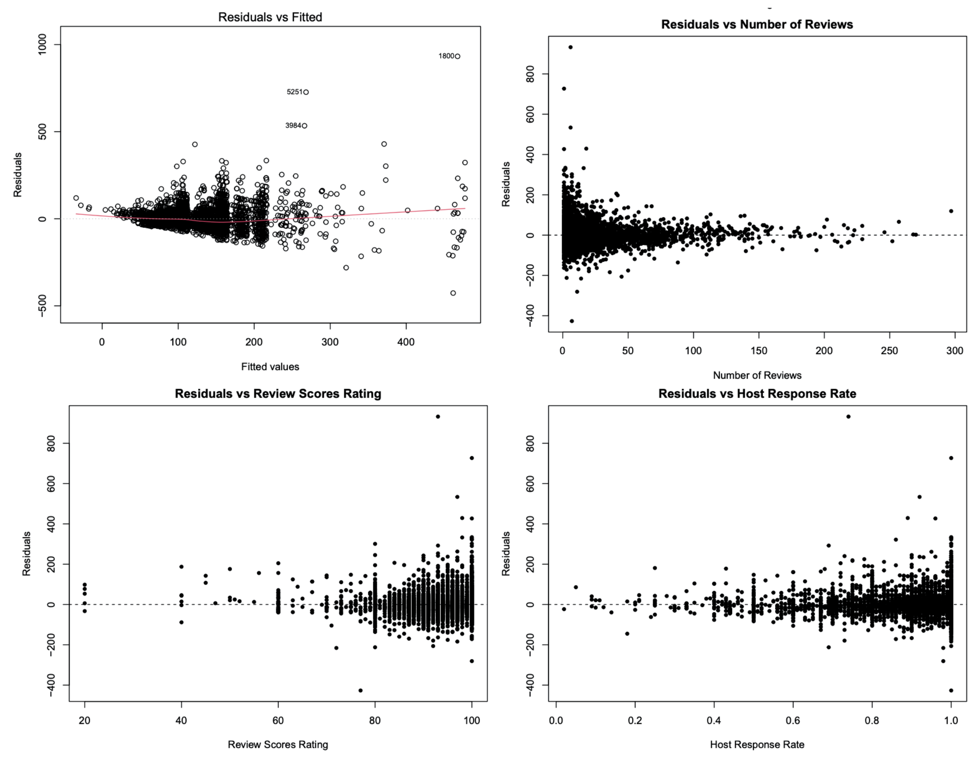
Motivation: It is not clear whether Airbnb hosts set prices using the same features that guests care about when choosing a stay. Understanding this alignment helps guests judge whether listings appear fairly priced, guides hosts toward pricing that better matches demand, and provides platforms and policymakers with evidence on how pricing actually works in practice.

Data Collection: We use the Amsterdam Airbnb listings dataset compiled by Dr. Philip E. Cannata (University of Texas at Austin, n.d.) and hosted on data. world. The file is derived from Inside Airbnb's public scrape of active Airbnb listings and was cleaned and curated in 2017 for educational use. This dataset is appropriate for our research question because it links each listing's price to detailed host, property, and review information, allowing us to study whether hosts' pricing decisions align with features that reflect consumer preferences.

Methods of Analysis



Original model: Price = Accommodates + Private_Room + Shared_Room + Accommodates:Private_Room + Accommodates:Shared_Room + Number_of_Reviews + Review_Scores_Rating + Number_of_Reviews:Review_Scores_Rating + Host_Response_Rate + hr_within_hour + hr_within_few_hours + hr_few_days_or_more



Tran_model: ln_price = Accommodates + Private_Room + Shared_Room + Accommodates:Private_Room + Accommodates:Shared_Room + ln_num_reviews + review_score_sq + ln_num_reviews: review_score_sq + host_response_sq + hr_within_hour + hr_within_few_hours + hr_few_days_or_more

model_auto: ln_price = Accommodates + Private_Room + Shared_Room + Accommodates:Private_Room + Accommodates:Shared_Room + ln_num_reviews + review_score_sq + hr_within_hour + hr_within_few_hours + hr_few_days_or_more + ln_num_reviews: review_score_sq

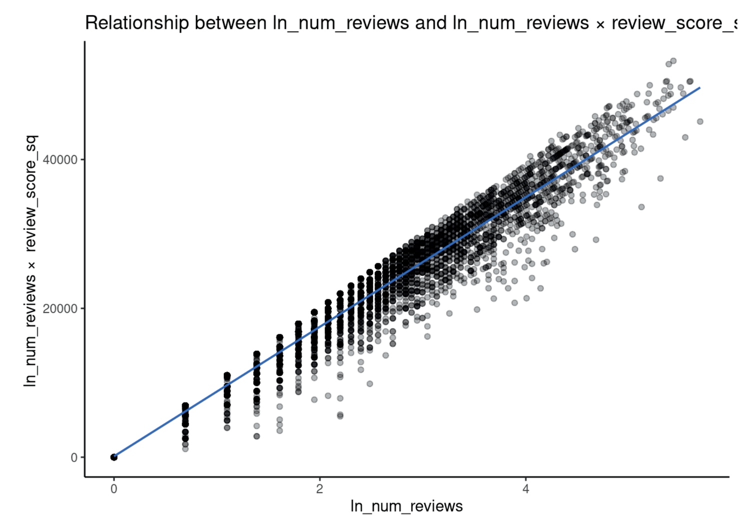
Nested F-test comparing transformed full model and Auto model

Comparison	Extra_term	df_num	df_den	F	p_value
Full vs Auto(reduced) model	ln_host_response	1	5586	0.022	0.882

Impact of multicollinearity on coefficient variance in the model_auto

Predictor	VIF
ln_num_reviews:review_score_sq	56.81
ln_num_reviews	56.05
Private_Room	4.14
Accommodates:Private_Room	3.84
review_score_sq	2.21
Accommodates:Shared_Room	1.71
Shared_Room	1.69
hr_within_hour	1.46
hr_within_few_hours	1.44
Accommodates	1.24
hr_few_days_or_more	1.07

Highly correlated → Delete the interaction



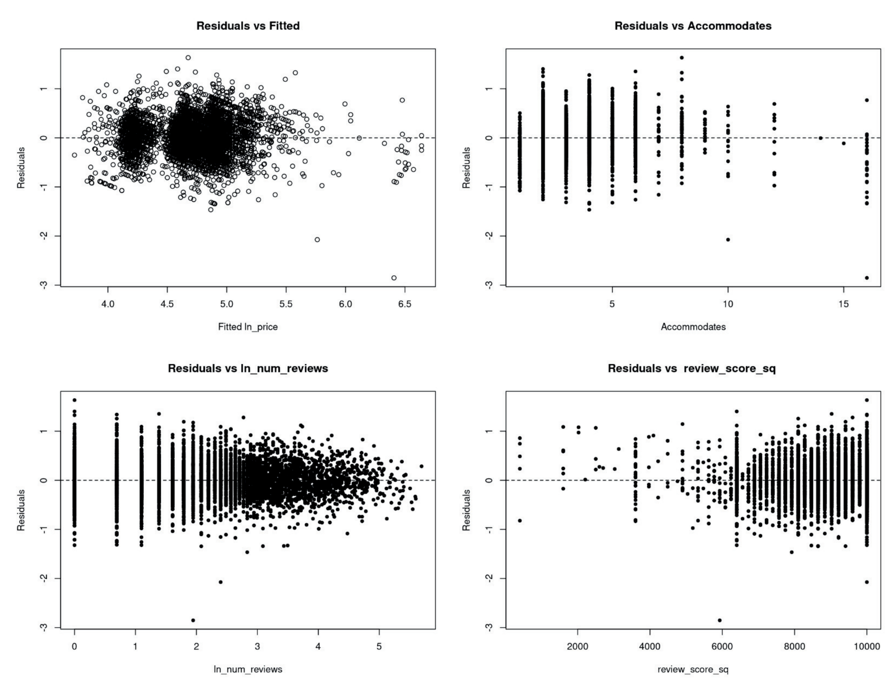
Impact of multicollinearity on coefficient variance in the model_final

Predictor	VIF
Private_Room	4.14
Accommodates:Private_Room	3.84
Accommodates:Shared_Room	1.71
Shared_Room	1.69
hr_within_hour	1.46
hr_within_few_hours	1.44
Accommodates	1.23
hr_few_days_or_more	1.07
review_score_sq	1.04
ln_num_reviews	1.03

Comparison of Tran_model and model_final

Model	Adj_R2	AIC	AICc	BIC
Tran_model	0.457	-11536.52	-11536.44	-11450.32
model_final	0.450	-11468.03	-11467.97	-11395.09

model_final: ln_price = Accommodates + Private_Room + Shared_Room + Accommodates:Private_Room + Accommodates:Shared_Room + ln_num_reviews + review_score_sq + hr_within_hour + hr_within_few_hours + hr_few_days_or_more



Ethics Discussion

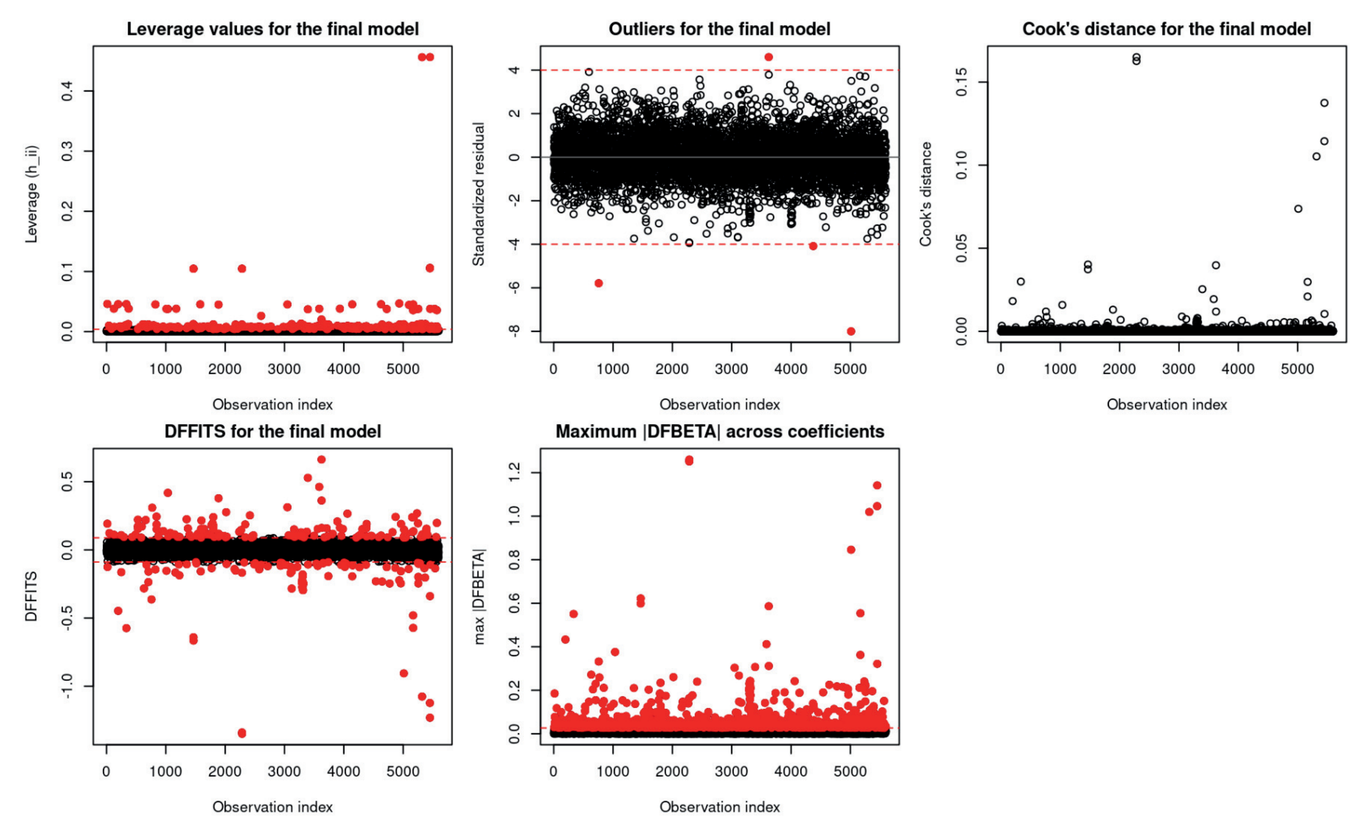
- Ignores context and purpose of the model
- Ignores the multicollinearity
- Ignores model violations
- Less blameworthy since manual recheck

	Estimate	Std.Error	t value	p value	Significant
(Intercept)	4.059	0.039	104.928	0.000	Significant
Accommodates	0.133	0.003	43.712	0.000	Significant
Private_Room	-0.229	0.025	-9.079	0.000	Significant
Shared_Room	-0.463	0.087	-5.331	0.000	Significant
Accommodates:Private_Room	-0.084	0.009	-9.410	0.000	Significant
Accommodates:Shared_Room	-0.128	0.018	-7.047	0.000	Significant
ln_num_reviews	-0.034	0.004	-8.764	0.000	Significant
review_score_sq	0.000	0.000	10.368	0.000	Significant
hr_within_hour	0.025	0.013	2.017	0.044	Significant
hr_within_few_hours	0.047	0.012	3.945	0.000	Significant
hr_few_days_or_more	-0.171	0.033	-5.243	0.000	Significant

Result

Conclusions

- "Entire home" has the largest positive coefficient, meaning whole homes are priced substantially higher, consistent with Wang & Jeong (2018), who find entire-home listings earn more. Positive coefficients for Accommodates and fast response-time categories show that listings hosting more guests and with more responsive hosts charge higher prices, matching the capacity and host–guest relationship effects reported in Wang & Jeong (2018) and Chang & Li (2021), respectively.
- However, ln(num_reviews) and review_score_sq have coefficients close or equal to 0, meaning listings with more reviews are slightly cheaper, which contrasts with Teubner et al. (2017), who found that more reviews are associated with higher prices.



Limitations

- Model Assumptions:
  - Violation of the homoscedasticity assumption, likely due to limited variation in review scores - most listings received similarly high ratings.
  - Violation of the uncorrelated assumption, likely due to model misspecification across different subgroups of listings
- Model Fit:
  - model_final: Adjusted R-squared is less than 50%
- Problematic Observations:
  - A few high-leverage and influential points slightly affect stability but not key predictors
- Data Scope:
  - Listing-level data only; missing host, neighborhood, and seasonal factors may introduce small bias
- Future Work:
  - Collect more diverse data to improve model fit and predictive accuracy

References: Cannata, P. E. (2017). GAirBnB [Data set]. data.world. https://data.world/cannata/gaairbnb

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